
Partition-Aware Message-Passing Complexity

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Abstract

It is shown that for any finite partition Π of a finite vertex set V and any binary task $f : V \rightarrow \{0, 1\}$, the cell-wise Bayes error ε_{Π}^* and the partition-conditional entropy $H(f \mid \Pi)$ (in bits) satisfy the two-sided sandwich

$$H_{\text{bin}}^{-1}(H(f \mid \Pi)) \leq \varepsilon_{\Pi}^* \leq \frac{1}{2} H(f \mid \Pi).$$

The upper bound is tight in the strong sense that no inequality $\varepsilon_{\Pi}^* \leq c H(f \mid \Pi)$ with $c < 1/2$ holds uniformly. The lower bound is tight at every $\varepsilon \in [0, 1/2]$ and recovers the Fano (1961) / Hellman–Raviv (1970) / Feder–Merhav (1994) entropy–error inequalities as the degenerate-partition case.

It is further shown that specialising Π to the partition $\Pi_{\mathcal{A}}(G, L)$ that a depth- L message-passing family $\mathcal{A}$ induces on $V(G)$ — *Partition-Aware Message-Passing Complexity* (PA-MPC) — yields a training-free, family-indexed two-sided error bracket comparable across GCN, GIN, GAT, GatedGCN, and GraphSAGE. The sandwich is verified on a finite-graph Lean 4 witness ($C_3, \dots, C_8$ and Petersen, zero `sorry`) and on a 1000-row exact-rational ledger. The accompanying continuous-transfer conjecture C1 is mathematically open; its empirical status was frozen on 2026-05-30 as *supported-by-evidence at $L \geq 3$* on the eight-graph non-vertex-transitive anchor.

1 Introduction

The expressivity of message-passing neural networks (MPNNs) has a well-developed structural ceiling: the 1-Weisfeiler–Leman (WL) discrimination power bounds what depth- L MPNNs can resolve about graphs (Xu et al., 2019; Morris et al., 2019). The partition-explicit message-passing complexity (MPC) framing of Kemper et al. (2025) refines this picture by quantifying, for the partition a model induces, how much task information remains.

Two practical problems remain.

1. **Architecture occlusion.** MPC depends on a particular model; its value cannot be compared meaningfully across families (GCN vs. GIN vs. GAT vs. GraphSAGE vs. GatedGCN) because the partitions those families *can* induce differ at depth 0 in ways MPC does not expose.
2. **No two-sided guarantee.** MPC says something about what the model resolves, but does not give a two-sided sandwich on predictability of the task by the family at that depth, independent of training.

PA-MPC addresses both. By indexing on the *family-level* initial partition $\Pi_{\mathcal{A}}^{(0)}$ and applying the same LossyWL operator to depth L , PA-MPC is well-defined before any training and is comparable across families. The bridge inequality of § 3 pins its operational meaning as a two-sided sandwich for the WL-determined task class $\mathcal{F}_{\text{WL}}$, up to one bit.

Organisation and contributions. The rest of the paper is organised as follows. § 2 positions PA-MPC against three immediate neighbours; § 3 develops the theory; § 4 summarises the empirical and mechanised verification; § 5 reports the frozen status of conjecture C1; §§ 6 and 7 close the paper. Our contributions are organised around four pillars.

- (a) *Theory*. Definitions, the two-sided bridge inequality (Theorem 8), refinement monotonicity (Theorem 9), architecture factorisation (Theorem 10), the PA-MPC instantiation (Theorem 11), sharpness (Theorem 12), the prior-aware Fano sharpening (Theorem 13), and scope.
- (b) *Lean witness*. A Lean 4 discrete bridge witness (PAMPC-LEAN-WITNESS-C4-K) on $\{C_4, \dots, C_8, \text{Petersen}\}$, zero `sorry`, trusted-kernel-only axiom base.
- (c) *Empirical*. Three exact-rational ledgers (E03, E07, E09; tier L-I), one float-tier oversmoothing baseline (E01), and three honestly-pilot results (E04, E06, E08).
- (d) *Reproducibility*. A sealed Merkle artefact store, a deterministic claim-manifest build pipeline, six decision gates, and methodology manuals recording exactly how every reported number was derived.

Status discipline. Throughout, we keep four statuses sharply separate.

- *Proved (claim-pinned)*. Theorem 8, Theorems 9 and 10, Theorem 11, Theorem 7.
- *Proved (paper-only, no claim ID)*. Theorem 12 (sharpness) and Theorem 13 (prior-aware) are hand proofs; they are **deliberately not** pinned to a trust tier (L-I / L-II / float) and not part of the claim manifest. Their correctness rests on the appendix proof alone; the L-I / L-II / float infrastructure is unaffected by either.
- *Mechanised / exact-audited*. Lean witness; exact- $\mathbb{Q}$ ledgers E02, E03, E07, E09 (L-I).
- *Empirical / conjectural*. Float baseline E01; pilot apparatus E04, E06, E08; the continuous-transfer conjecture C1 (open; empirically frozen 2026-05-30).

2 Related Work

We position PA-MPC against three immediate neighbours. An extended discussion of antecedents (ergodic theory, algorithmic partition refinement, oversmoothing and over-squashing, the 2022–2024 partition-indexed GNN wave, and learning-theoretic uses of refinement-monotonicity) is deferred to § B.

Expressivity ceiling. Xu et al. (2019) and Morris et al. (2019) show that MPNNs are at most as expressive as 1-WL on graph discrimination, and Azizian & Lelarge (2021) sharpen this to an exact characterisation. PA-MPC is orthogonal: we *quantify* predictability of a fixed task on a fixed graph rather than characterise isomorphism power. Higher-order k -WL (Morris et al., 2020) is explicitly out of scope; § 3.3 gives the exclusion criterion.

Partition-explicit MPC. Kemper et al. (2025) frame complexity by the partition a single trained model induces. PA-MPC lifts that framing to an *architecture-family / initial-observable pair* $(\mathcal{A}, \Pi^{(0)})$ — well-defined before training and comparable across MPNN families — and packages it with a two-sided Bayes-error sandwich (Theorem 8).

Equitable partitions. The $L \rightarrow \infty$ limit of $\Pi_{\mathcal{A}}(G, L)$ for any 1-WL-template family is the equitable partition of G (Schwenk, 1974; Godsil & Royle, 2001). That object is the most direct intellectual ancestor of Theorem 2. The novelty here is the family indexing (Theorem 10) and the Bayes-error packaging (Theorem 8), not the partition itself.

3 Theory

3.1 Definitions

Remark 1 (finite, measure-free setting). Throughout, V is finite and we write $\mathbb{P}[\cdot] := |\{v : \cdot\}|/|V|$. We use no σ -algebras, no countable additivity, no dominated convergence, no Radon–Nikodým arguments.

Vertex weighting is uniform, $q_v := 1/|V|$; non-uniform weighting amounts to replacing the counting normaliser by a positive weight summing to 1, with no other change in any proof.

Definition 2 (admissible partition family $\Pi_{\mathcal{A}}$). Let $G = (V, E)$ be a finite graph and let $\mathcal{A}$ be a depth- L MPNN architecture family with declared initial-observable partition $\Pi_{\mathcal{A}}^{(0)}$. The *depth- ℓ LossyWL partition* of $\mathcal{A}$ on G is $\Pi_{\mathcal{A}}^{(\ell)} := \text{LossyWL}^{\ell}(\Pi_{\mathcal{A}}^{(0)}; G)$ where LossyWL refines a partition by appending the multiset of neighbour-cell labels to each vertex’s label and reducing. $\Pi_{\mathcal{A}}(G, L) := \Pi_{\mathcal{A}}^{(L)}$.

Definition 3 (partition-conditional entropy). For a binary task $f : V \rightarrow \{0, 1\}$ and a finite partition Π of V with cells C , masses $q_C := |C|/|V|$, and posteriors $P_C := \mathbb{P}[f = 1 \mid C]$,

$$H(f \mid \Pi) := \sum_{C \in \Pi} q_C \cdot H_{\text{bin}}(P_C) = \mathbb{E}[H_{\text{bin}}(P_{C(v)})],$$

where $H_{\text{bin}}(p) := -p \log_2 p - (1 - p) \log_2 (1 - p)$. Both forms vanish iff Π is *pure* for f (Theorem 7).

Definition 4 (PA-MPC). $\text{PA-MPC}(f, G; \mathcal{A}, L) := H(f \mid \Pi_{\mathcal{A}}(G, L))$.

Definition 5 (WL-determined task class). $\mathcal{F}_{\text{WL}}(G)$ is the set of tasks constant on every cell of the WL-stable partition $\Pi^{\text{WL}}(G) := \text{LossyWL}^{\infty}(\Pi^{(0)}; G)$. “Determined” rather than “measurable”: the condition is set-theoretic — f factors through $V \rightarrow \Pi^{\text{WL}}$ — and carries no measure-theoretic content.

Definition 6 (admissible architecture family). A pair $(\mathcal{A}, \Pi_{\mathcal{A}}^{(0)})$ is an *admissible* depth- L architecture-family specification if every member $h \in \mathcal{A}_L$ satisfies the following four conditions.

- (1) The depth-0 hidden state $h^{(0)}(v)$ depends on v only through the cell of v in $\Pi_{\mathcal{A}}^{(0)}$.
- (2) There is a single update map ϕ_{ℓ} at each layer ℓ , applied identically at every vertex.
- (3) ϕ_{ℓ} acts on the *multiset* of neighbour states.
- (4) Beyond $\Pi_{\mathcal{A}}^{(0)}$, ϕ_{ℓ} has no access to vertex names or any feature not already determined by the current colour class.

GCN, GIN, GatedGCN, GAT (with permutation-invariant attention pooling), and GraphSAGE all admit specifications meeting (1)–(4); higher-order k -WL variants are out of scope (§ 3.3).

Lemma 7 (purity $\Leftrightarrow$ zero conditional entropy). *Call Π pure for f if $P_C \in \{0, 1\}$ for every $C \in \Pi$ — equivalently, f is constant on every cell of Π , i.e. f factors through the quotient map $V \rightarrow V/\Pi$. Then*

$$\text{Pure}(f, \Pi) \iff H(f \mid \Pi) = 0 \iff \mathbb{E}[\text{Var}(f \mid \Pi)] = 0.$$

Proof. Each summand $q_C \cdot H_{\text{bin}}(P_C)$ is non-negative and vanishes iff $P_C \in \{0, 1\}$; the $P_C \in \{0, 1\}$ condition says f is constant on C (it equals its mean). The same argument applies to $\mathbb{E}[\text{Var}(f \mid \Pi)]$ via $P_C(1 - P_C) \geq 0$. $\square$

Naming. The condition “ f is constant on every cell of Π ” has several established names across nearby literatures, all equivalent on a finite vertex set: Π **resolves** f (partition-lattice combinatorics; $\ker f$ is coarsened by Π); **f -equitable partition** (algebraic graph theory: Schwenk, 1974; Godsil & Royle, 2001); **pure partition** (decision-tree induction: Breiman et al., 1984); **zero-equivocation partition** (Shannon, 1948).

We use *pure* as the display term — the decision-tree mnemonic is the closest match for a binary task — but readers from any of these lineages may substitute their own term verbatim; nothing in the proofs depends on the choice.

3.2 The Partition Bayes-Entropy Sandwich

Partition Bayes error. For a finite partition Π of V and a binary task f , define

$$\varepsilon_{\Pi}^* := \min_{g: \Pi \rightarrow \{0, 1\}} \mathbb{P}[g(C(v)) \neq f(v)] = \sum_{C \in \Pi} q_C \cdot \min(P_C, 1 - P_C),$$

attained by the plug-in rule $\hat{h}_{\Pi}(v) := \mathbf{1}\{P_{C(v)} \geq 1/2\}$. Equivalently, ε_{Π}^* is the Bayes risk of f when the predictor is constrained to be constant on each cell of Π (Devroye et al., 1996, §2.1, recast for finite V).

Theorem 8 (partition Bayes-entropy sandwich). *Let Π be a finite partition of a finite vertex set V and let $f : V \rightarrow \{0, 1\}$ be any binary task. Then*

$$H_{\text{bin}}^{-1}(H(f | \Pi)) \leq \varepsilon_{\Pi}^* \leq \frac{1}{2} H(f | \Pi),$$

where $H_{\text{bin}}^{-1} : [0, 1] \rightarrow [0, 1/2]$ is the inverse of binary entropy on the increasing branch.

The theorem makes no reference to G , $\mathcal{A}$, the LossyWL operator, or $\mathcal{F}_{\text{WL}}$. It is a property of any finite partition and any binary labelling. (Upper side: Hellman & Raviv, 1970. Lower side: Fano, 1961 in its binary form, sharpened by Feder & Merhav, 1994.) The full proof is in § A; we sketch the two halves as the two extremal directions of a single Lagrangian.

Single program, two directions. Let $e_C := \min(P_C, 1 - P_C) \in [0, 1/2]$ and use the symmetry $H_{\text{bin}}(P_C) = H_{\text{bin}}(e_C)$, so that $H(f | \Pi) = \sum_C q_C H_{\text{bin}}(e_C)$ and $\varepsilon_{\Pi}^* = \sum_C q_C e_C$. Both bounds are optimal values of opposite Jaynes max-entropy programs on $\{(q_C, e_C)\}_C$:

$$\begin{aligned} \text{(Fano lower)} \quad & \max_{\{q_C, e_C\}} \sum_C q_C H_{\text{bin}}(e_C) \text{ s.t. } \sum_C q_C e_C = \varepsilon \implies e_C \equiv \varepsilon, \\ \text{(HR upper)} \quad & \max_{\{q_C, e_C\}} \sum_C q_C e_C \text{ s.t. } \sum_C q_C H_{\text{bin}}(e_C) = H \implies e_C \in \{0, 1/2\}. \end{aligned}$$

The Fano direction places the optimiser at the unique interior critical point of H_{bin} 's concavity, giving $H \leq H_{\text{bin}}(\varepsilon)$ and, on inversion, $H_{\text{bin}}^{-1}(H) \leq \varepsilon_{\Pi}^*$. The Hellman–Raviv direction places the optimiser at the boundary of $[0, 1/2]$, where $e \leq \frac{1}{2} H_{\text{bin}}(e)$ is tight, giving $\varepsilon_{\Pi}^* \leq \frac{1}{2} H(f | \Pi)$.

Achievable region and slack. Every finite partition and binary task on a finite vertex set realises a pair $(\varepsilon_{\Pi}^*, H(f | \Pi))$ in $\tilde{A}_2 := \{(\varepsilon, H) \in [0, 1/2] \times [0, 1] : H_{\text{bin}}^{-1}(H) \leq \varepsilon \leq H/2\}$. The width $w(H) := \frac{1}{2}H - H_{\text{bin}}^{-1}(H)$ vanishes at $H \in \{0, 1\}$ and attains its maximum $w^* \approx 0.161$ at $\varepsilon = 1/5$; the sandwich therefore localises ε_{Π}^* to a window of width at most ≈ 0.161 uniformly in Π and f .

Proposition 9 (refinement monotonicity). *If Π' refines Π (every cell of Π' is contained in some cell of Π), then for every binary f ,*

$$H(f | \Pi') \leq H(f | \Pi), \quad \varepsilon_{\Pi'}^* \leq \varepsilon_{\Pi}^*.$$

Sketch. Each $C \in \Pi$ partitions into $C = \bigsqcup_j C'_j$ with masses $q_{C'_j}$ summing to q_C and posteriors $P_{C'_j}$ averaging (weights $q_{C'_j}/q_C$) to P_C . Concavity of H_{bin} gives $\sum_j q_{C'_j} H_{\text{bin}}(P_{C'_j}) \leq q_C H_{\text{bin}}(P_C)$; sum over C . The same argument applies to $\min(p, 1 - p)$ via its concavity on $[0, 1]$. Full proof in § A. $\square$

Proposition 10 (architecture factorisation). *Let $(\mathcal{A}, \Pi_{\mathcal{A}}^{(0)})$ be admissible (Theorem 6) and let $\Pi = \Pi_{\mathcal{A}}(G, L)$. Then every depth- L predictor $h \in \mathcal{A}_L$ is constant on every cell of Π .*

Proof. Induction on ℓ : the layer- ℓ hidden state $h^{(\ell)}(v)$ depends on v only through the cell of v in $\Pi_{\mathcal{A}}^{(\ell)}$. Base: Theorem 6(1). Step: if v, w share a cell of $\Pi_{\mathcal{A}}^{(\ell+1)}$ then they share a cell of $\Pi_{\mathcal{A}}^{(\ell)}$ and the multisets of neighbour-cell labels coincide; by induction, the multisets of neighbour states coincide; Theorem 6(2–4) gives $h^{(\ell+1)}(v) = h^{(\ell+1)}(w)$. $\square$

Corollary 11 (PA-MPC sandwich for admissible families). *Let $(\mathcal{A}, \Pi_{\mathcal{A}}^{(0)})$ be admissible, let G be a finite graph, $f : V \rightarrow \{0, 1\}$ any binary task, $\Pi = \Pi_{\mathcal{A}}(G, L)$, and PA-MPC = $H(f | \Pi)$. Then: (i) every decision rule $\hat{F} = \tau \circ h$ with $h \in \mathcal{A}_L$ has error $\geq \varepsilon_{\Pi}^* \geq H_{\text{bin}}^{-1}(\text{PA-MPC})$; (ii) the plug-in $\hat{h}_{\Pi}$ realises an admissible-family predictor with error $\varepsilon_{\Pi}^* \leq \frac{1}{2} \text{PA-MPC}$; (iii) increasing depth $L \rightarrow L + 1$ refines Π and so cannot increase PA-MPC or ε^* ; (iv) if $f \in \mathcal{F}_{\text{WL}}(G)$, the bound is non-vacuous in the limit: $\lim_{L \rightarrow \infty} \text{PA-MPC} = 0$.*

Proof. (i) combines Theorem 10 with the lower bound of Theorem 8; (ii) is the upper bound; (iii) is Theorem 9; (iv) holds because $f \in \mathcal{F}_{\text{WL}}(G)$ is constant on the WL-stable partition. $\square$

Operational reading. PA-MPC sandwiches the irreducible partition Bayes error of depth- L message passing on both sides — no additive bit-slack, no log-axis ambiguity.

PA-MPC = 0 iff the partition is pure; positive PA-MPC forces positive ε_Π^* via the lower bound; small PA-MPC forces small ε_Π^* via the upper bound; deeper architectures never increase the bound.

Proposition 12 (no closed-form improvement of the sandwich). *Both bounds in Theorem 8 are uniformly unimprovable in the following sense.*

- (i) *For every constant $c < 1/2$ there exists a finite partition Π and binary task f on finite V with $\varepsilon_\Pi^* > c \cdot H(f \mid \Pi)$.*
- (ii) *For every continuous $\psi : [0, 1] \rightarrow [0, 1/2]$ with $\psi(H_0) > H_{\text{bin}}^{-1}(H_0)$ at some $H_0 \in (0, 1)$, there exist Π and f with $\varepsilon_\Pi^* < \psi(H(f \mid \Pi))$.*

Proof. (i) The Hellman–Raviv witnessing family Π_α^{HR} (§ A, Tightness) achieves $\varepsilon_{\Pi_\alpha^{\text{HR}}}^* = \frac{1}{2}H(f \mid \Pi_\alpha^{\text{HR}})$ for every $\alpha \in (0, 1]$. (ii) By continuity, $\psi(H) > H_{\text{bin}}^{-1}(H)$ on a neighbourhood of H_0 ; pick $\varepsilon \in \mathbb{Q} \cap (0, 1/2)$ with $H_{\text{bin}}(\varepsilon)$ in that neighbourhood and apply the Fano witnessing family at any n with $\varepsilon n \in \mathbb{Z}$. $\square$

The rhetorical content of Theorem 12 is Massey-style (Massey, 1994, §III): a positive sharpness claim upgraded to a negative non-existence claim. The achievable region A_2 is the tightest two-sided closed-form bracket on $(\varepsilon_\Pi^*, H(f \mid \Pi))$ available in the binary case.

Proposition 13 (prior-aware sharpening). *Let Π be a finite partition of finite V and $f : V \rightarrow \{0, 1\}$ a binary task. With marginal frequency $P_f := |\{v : f(v) = 1\}|/|V|$, marginal Bayes error $\varepsilon_\emptyset^* := \min(P_f, 1 - P_f)$, and mutual information $I(f; \Pi) := H_{\text{bin}}(P_f) - H(f \mid \Pi)$,*

$$d_{\text{KL}}(\varepsilon_\Pi^* \parallel \varepsilon_\emptyset^*) \leq I(f; \Pi),$$

where $d_{\text{KL}}(p \parallel q) := p \log_2 \frac{p}{q} + (1 - p) \log_2 \frac{1-p}{1-q}$.

Proof. § A, equations (A.10)–(A.12), via data-processing on the binary error indicator (Han & Verdú, 1994). $\square$

Status (trust-tier note). Theorem 13 is a **paper-only hand proof**. It carries no PAMPC-* claim ID, is not part of the L-I exact-rational ledger, not part of the L-II Lean witness, and not behind any G0..G4 decision gate.

Theorem 13 is moreover *out of scope* for the Lean chain and the exact-rational ledger by design: it will not be mechanised in `PaMpc.BayesErrorBridge` and no E02 row will be added for it in this programme. Any prior-aware result that needs Lean coverage must be introduced as a separate claim ID (e.g. PAMPC-PRIOR-*) in a successor paper, not as a retrofit to the existing bridge module.

Substituting the trivial bound $\varepsilon_\emptyset^* = 1/2$ in Theorem 13 recovers exactly Theorem 8’s lower bound; for skewed marginals the bound is strictly tighter.

3.3 Scope

The partition theorem (Theorem 8) and refinement proposition (Theorem 9) are stated for *any* finite partition and *any* binary task on a finite vertex set; nothing about graphs, WL, or admissible architectures enters those statements.

The architecture factorisation (Theorem 10) and the PA-MPC sandwich (Theorem 11) require the admissibility conditions of Theorem 6. The non-vacuity clause Theorem 11(iv) is the one place where $f \in \mathcal{F}_{\text{WL}}(G)$ is used.

Higher-order WL templates (Morris et al., 2020), which violate admissibility (4) by referencing k -tuples, are out of scope. Multi-class extensions ($|\mathcal{Y}| > 2$) are routine via the concave-envelope construction of Feder & Merhav (1994) and are deferred to follow-up work.

3.4 Three named conjectures (registry)

Table 1: Conjecture registry. C1 is the only conjecture whose resolution gates a phase transition in the programme; its precise statement, hypotheses H1–H3, decision rule, and sealed empirical verdict are in § 5.

ID	Title	Falsifier
C1	Continuous-transfer (open; frozen 2026-05-30)	E08 (§ 5)
C2	K_B rank saturation on cycles	E03
C3	Feature-richness boundary	E07

3.5 Remark on $MI^2 \approx 1/2$

Under purity, $MI^2(f; \Pi) = \frac{1}{2}R^2(f; \Pi) = \frac{1}{2}$ regardless of cell count — operationally vacuous in the solvable regime. The partition-conditional entropy is the *informative dual*: where MI^2 becomes constant, the conditional entropy vanishes; before that point, it carries the architecture-level signal.

4 Empirical and Mechanised Verification (summary)

The full experimental record (E01–E09, per-experiment LAB-PLANS, exact-rational ledger contents, Lean module listings, and the reproducibility infrastructure) is held in the project’s Markdown companion (PAPER-ARXIV.md, §4–§8) and the 20-experiments/ tree. We summarise the headline outcomes.

Exact-rational ledger (tier L-I). Three sealed ledgers verify the theory on a synthetic anchor with L-I-computable WL partition and exact- $\mathbb{Q}$ partition-conditional entropy. **E03** (audit of Theorem 9): 280 rows ($\{P_4, \dots, P_{12}, \text{Petersen}\} \times 4 \text{ tasks} \times 7 \text{ depths}$); all three falsifiers PASS (monotonicity, purity on vertex-transitive, WL-stability scaling). **E07** (feature-richness boundary): GIN- vs. GCN-archetype init on the same grid; refinement chain holds on 280/280 rows, 18 strict-improvement rows witness Conjecture C3 non-vacuous, zero regressions. **E09** (intervention pricing): rewiring price ≥ 0 on WL-cell tasks, all invariants PASS; sign-variability is preserved as an honest non-gating advisory.

Lean witness (tier L-II). A Lean 4 module verifies the discrete bridge on $\{C_4, \dots, C_8, \text{Petersen}\}$ at depths 0..3 via `native_decide`; zero `sorry`; trusted-kernel-only axiom base (`Classical.choice`, `propext`). Claim PAMPC-LEAN-WITNESS-C4-K.

Float-tier baseline. **E01** (oversmoothing): lazy random walk on bipartite anchors reproduces depth-induced collapse qualitatively (smoothed-window F2). **E04**, **E06**, **E08** are honestly pilot.

Reproducibility skeleton. Sealed Merkle artefact store (10 experiments / 27 leaves); deterministic claim-manifest build pipeline; six decision gates (G0..G4 PASS).

5 Continuous-Transfer Conjecture C1 (empirical status)

C1 (continuous-transfer; mathematically open). For every admissible MPNN family $\mathcal{A}$ and every binary task $f \in \mathcal{F}_{\text{WL}}(G)$, the depth-monotone PA-MPC trajectory $L \mapsto \text{PA-MPC}(f, G; \mathcal{A}, L)$ admits a continuous L -extension on a sealed substrate, with stage-resolved decay rates that depend only on the architectural class $\{\text{GCN}, \text{GIN}, \text{GatedGCN}\}$ vs. $\{\text{GAT}, \text{GraphSAGE}\}$.

Empirical verdict (frozen 2026-05-30). Under the literal §9 protocol of PAPER-ARXIV.md, status *supported-by-evidence at $L \geq 3$* on the eight-graph non-vertex-transitive anchor anchors, with the architectural dichotomy $\{\text{GCN}, \text{GIN}, \text{GatedGCN}\} / \{\text{GAT}, \text{GraphSAGE}\}$. The companion conjecture C1’ (2-WL substrate, E10) is the headline open problem for paper-02.

6 Known Limitations and Honest Narrowings

1. **E06 demoted** from headline to inconclusive-pilot.
2. **E09 F3 reformulated** from sign-variability to “rewiring price ≥ 0 on WL-cell tasks”.
3. **E01 required three task/operator redesigns.**
4. **G4 narrowed** from PDF to claim manifest (byte-determinism without strict SOURCE_DATE_EPOCH).
5. **No mode=full for E04 and E08.** Sealed Stage K/L/M C1 verdict only.
6. **No real-world graph results.** ZINC/OGB/TUDataset are in scope for paper-02; this paper is theory plus the synthetic anchor.
7. **Theorems 12 and 13 are paper-only.** Both are hand proofs in § A and **by design** carry no PAMPC-* claim ID, no trust tier, and no decision gate. Theorem 13 is moreover *out of scope* for the Lean chain and the exact-rational ledger: it will not be mechanised in `PaMpc.BayesErrorBridge` and no E02 row will be added for it in this programme. Any prior-aware result that needs Lean coverage must be introduced as a separate claim ID (e.g. `PAMPC-PRIOR-*`) in a successor paper, not as a retrofit to the existing bridge module. The L-I / L-II / float infrastructure is thereby a closed system with respect to either proposition.

7 Conclusion

PA-MPC is a partition-explicit, architecture-family-indexed generalisation of MPC. The discrete bridge inequality (Theorem 8) pins its operational meaning as a two-sided sandwich on WL-determined predictability. The exact-rational ledger and the Lean L-II witness verify the theory on a synthetic anchor.

The continuous-transfer conjecture C1 is frozen as of 2026-05-30: supported-by-evidence at $L \geq 3$ on `anchors`, with the architectural dichotomy $\{\text{GCN}, \text{GIN}, \text{GatedGCN}\}$ versus $\{\text{GAT}, \text{GraphSAGE}\}$. The companion conjecture C1’ is registered as the headline open problem for paper-02.

The reproducibility skeleton is closed end-to-end. The kill branch of C1 is publishable; the PASS branch is publishable; and the honest pilot status of E04, E06, E08 is publishable.

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A Proof of Theorem 8

Setup. Fix a finite vertex set V , a binary task $f : V \rightarrow \{0, 1\}$, and a finite partition Π of V with cells C , masses $q_C := |C|/|V|$, and posteriors $P_C := \mathbb{P}[f = 1 \mid C]$. By Remark 1, vertex weighting is uniform, $q_v := 1/|V|$.

Upper bound (Hellman–Raviv). For each cell, let $e_C := \min(P_C, 1 - P_C) \in [0, 1/2]$. The pointwise inequality $e \leq \frac{1}{2} H_{\text{bin}}(e)$ on $[0, 1/2]$ (Hellman & Raviv, 1970) gives, summing weighted by q_C and using the symmetry $H_{\text{bin}}(P_C) = H_{\text{bin}}(e_C)$,

$$\varepsilon_{\Pi}^* = \sum_C q_C e_C \leq \frac{1}{2} \sum_C q_C H_{\text{bin}}(e_C) = \frac{1}{2} H(f \mid \Pi). \quad (\text{A.1–A.2})$$

Lower bound (Fano, binary form). The function H_{bin} is concave on $[0, 1/2]$. Jensen’s inequality applied to the cell-mass distribution $\{q_C\}$ and the values $\{e_C\}$ yields

$$H(f \mid \Pi) = \sum_C q_C H_{\text{bin}}(e_C) \leq H_{\text{bin}}(\sum_C q_C e_C) = H_{\text{bin}}(\varepsilon_{\Pi}^*). \quad (\text{A.3–A.5})$$

Since H_{bin} is monotone-increasing on $[0, 1/2]$ with inverse $H_{\text{bin}}^{-1} : [0, 1] \rightarrow [0, 1/2]$,

$$H_{\text{bin}}^{-1}(H(f \mid \Pi)) \leq \varepsilon_{\Pi}^*. \quad (\text{A.6})$$

Combining (A.2) and (A.6) gives Theorem 8.

Proof of Proposition 9. If Π' refines Π , each $C \in \Pi$ partitions as $C = \bigsqcup_j C'_j$ with $q_{C'_j}$ summing to q_C . The cell posterior P_C equals the $q_{C'_j}$ -weighted average of $\{P_{C'_j}\}_j$, so by concavity of H_{bin} , $\sum_j q_{C'_j} H_{\text{bin}}(P_{C'_j}) \leq q_C H_{\text{bin}}(P_C)$. Summing over C gives $H(f \mid \Pi') \leq H(f \mid \Pi)$. The same argument with the concave $p \mapsto \min(p, 1 - p)$ on $[0, 1]$ gives $\varepsilon_{\Pi'}^* \leq \varepsilon_{\Pi}^*$. $\square$

Proof of Proposition 13 (prior-aware sharpening). Let $P_f := |\{v : f(v) = 1\}|/|V|$ and $\varepsilon_{\emptyset}^* := \min(P_f, 1 - P_f)$. Define the binary error indicator $Z(v) := \mathbf{1}\{\hat{h}_{\Pi}(v) \neq f(v)\}$ for the plug-in rule $\hat{h}_{\Pi}$ of §3.2. By construction $\mathbb{P}[Z = 1] = \varepsilon_{\Pi}^*$. Since Z is a deterministic function of (f, Π) , the finite-alphabet data-processing inequality (Cover & Thomas, 2006, Thm 2.8.1, specialised to discrete distributions — no measure-theoretic generality needed) on the Markov chain $f \rightarrow \Pi \rightarrow Z$ gives

$$I(f; Z) \leq I(f; \Pi). \quad (\text{A.10})$$

The joint distribution of (f, Z) has marginals P_f and ε_{Π}^* and is fully determined by $\mathbb{P}[f \neq \hat{h}_{\Pi}] = \varepsilon_{\Pi}^*$. Among all couplings of (f, Z) with these marginals, the minimum mutual information is attained when $\hat{h}_{\Pi}$ predicts the marginal mode and makes errors only on the minority; this minimum equals $d_{\text{KL}}(\varepsilon_{\Pi}^* \parallel \varepsilon_{\emptyset}^*)$ (Han & Verdú, 1994, eq. (6)). Hence

$$I(f; Z) \geq d_{\text{KL}}(\varepsilon_{\Pi}^* \parallel \varepsilon_{\emptyset}^*). \quad (\text{A.11})$$

Combining (A.10)–(A.11),

$$d_{\text{KL}}(\varepsilon_{\Pi}^* \parallel \varepsilon_{\emptyset}^*) \leq I(f; \Pi). \quad (\text{A.12})$$

Theorem 8’s lower bound is the special case $P_f = 1/2$ (then $\varepsilon_{\emptyset}^* = 1/2$ and $d_{\text{KL}}(\varepsilon \parallel 1/2) = 1 - H_{\text{bin}}(\varepsilon)$, so (A.12) reduces to $H(f \mid \Pi) \leq H_{\text{bin}}(\varepsilon_{\Pi}^*)$, equation (A.6)). For $P_f \neq 1/2$ the bound is strictly tighter. $\square$

Tightness. Both boundaries of $\tilde{A}_2$ are realised by explicit one-parameter families on finite V (uniform vertex weighting, Remark 1). *Fano (lower) boundary* $\Pi_{\varepsilon}^{\text{F}}$: for $\varepsilon \in [0, 1/2]$ and n with $\varepsilon n \in \mathbb{Z}$, take $\Pi_{\varepsilon}^{\text{F}} := \{V\}$ and let f have exactly εn ones; then $P = \varepsilon = e$ and Jensen is tight. *Hellman–Raviv (upper) boundary* Π_{α}^{HR} : take two cells of equal mass with $(P_1, P_2) = (1/2, 1/2 - \alpha)$ so that $e_1 = 1/2$, $e_2 = \alpha$ and the pointwise inequality $e \leq \frac{1}{2} H_{\text{bin}}(e)$ is tight at both endpoints.

B Extended Related Work

This appendix collects context that §2 only gestures at. None of it is load-bearing for the proofs or for the empirical claims.

Oversmoothing and over-squashing. Cai & Wang (2020) and Oono & Suzuki (2020) characterise depth-induced embedding collapse on bipartite graphs; our float-tier experiment E01 reproduces this qualitatively. Alon & Yahav (2021) frame over-squashing as an information-bottleneck phenomenon. The bridge inequality is agnostic to bottleneck architecture: it holds for any MPNN computing via depth- L rooted coloured neighbourhood aggregation.

Partition-refinement antecedents. (i) *Ergodic theory.* Kolmogorov (1958), Sinai (1959), and Rokhlin (1967) defined entropy as a supremum over refining partitions; refinement-monotonicity of $H(\Pi)$ is the structural axiom of that construction. Proposition 9 is the finite, combinatorial cousin (no σ -additivity used — see Remark 1). (ii) *Algorithmic partition refinement.* Hopcroft (1971) and Paige & Tarjan (1987) gave the algorithmic schema; 1-WL is the graph specialisation, used in McKay’s *nauty* (McKay, 1981) for isomorphism testing and sharpened into the k -WL hierarchy by Cai et al. (1992) and Grohe (2017). (iii) *Algebraic graph theory.* The equitable-partition ancestry is already foregrounded in §2; Schwenk (1974) and Godsil & Royle (2001, ch. 9) are the direct sources for the $L \rightarrow \infty$ fixpoint statement.

Modern partition-indexed GNN theory (2022–2024). Geerts & Reutter (2022) characterise MPNN expressivity by per-layer partition refinement; Morris et al. (2023) survey the field with equitable partitions as the WL fixpoint; Grohe (2021) connects partition refinement to counting-logic; Böker et al. (2023) define a metric on partitions and quantify how close MPNNs come to separating points — the closest existing work in spirit to our Bayes-error packaging. None of this wave delivers all three of (a) a two-sided Bayes-error bracket indexed by $\Pi_{\mathcal{A}}(G, L)$, (b) an exact-rational ledger on small graphs, and (c) a mechanised statement; that triple is the unique contribution of PA-MPC.

Learning-theoretic uses of refinement-monotonicity. Outside GNNs, Steinke & Zakynthinou (2020) and Asadi et al. (2018) apply refinement-monotonicity arguments to partitions of hypothesis classes for generalisation bounds — the same Proposition 9 move applied in learning theory.